

Extraction and Analysis of Character Interactions

Team Project — Character Interactions in Novels

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Abstract

This project builds a pipeline to extract characters and their interactions from a novel (Jane Austen’s *Pride and Prejudice*, public-domain text). The output is a list of canonical characters (with aliases) and a sentence-indexed list of interactions of the form (`sentence_index`, `character1`, `character2`, ...). We combined multiple NER tools (spaCy, Stanza, GLiNER), alias clustering, heuristic and coreference-based interaction detection and evaluated various methods, including local LLM analysis, sentiment analysis, and speaker detection. This report documents methods, implementation, quantitative comparison and qualitative analysis of results.

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1 Introduction

The goal of the project is to automatically extract characters and detect interactions between them in a novel. The final deliverable is a machine-readable representation of interactions as tuples with a timestamp (sentence index) and the set of participating characters. The approach prioritises reproducibility and explainability: each step is scripted and available in the project repository.

2 Related Work

Detecting character interactions can be a great way to more deeply understand the dynamics between characters in a book. It can result in a nice graph visualization, as seen with the “A Song of Ice and Fire” novels and “Game of Thrones” series¹ or with various movies².

There are many papers on this topic, from which we found one survey that captures all important aspects in detail. The rest of this section is based on this paper³.

Work on this topic can be divided into manual and automatic approaches, the latter being more popular as the technology developed. To extract characters, one should look for proper nouns (e.g., “Elizabeth Bennet”), pronouns (e.g., “she”), and nominals (e.g., “her eldest sister”). Much research uses off-the-shelf NER tools and takes only the Person entities from them. For alias resolution, string similarity and gender compatibility are used against character occurrences. Finally, interactions can be defined as either appearing in the same unit of time, having a direct verbal interaction, mentioning some character, direct interaction, or a combination of these. The methods are various and highly dependent on the definition of the interaction.

3 Task and data

We chose Jane Austen’s *Pride and Prejudice* (Project Gutenberg public domain text). The raw text is preprocessed with `prepare_book.py` to remove Gutenberg boilerplate and illustration blocks; the cleaned text is stored at:

- `data/pride_and_prejudice.txt` (raw)
- `data/pride_and_prejudice_cleaned.txt` (cleaned)

Manual curation note. After the automatic extraction and alias-resolution steps, the candidate list of characters was manually reviewed and corrected. The curated canonical list is provided in `results/character_extraction/correct_character_list.txt` and was used as the authoritative reference for the final analyses reported here. This manual quality-control step ensured removal or correction of clear noise and ambiguous entries before finalizing outputs.

4 Pipeline overview

The implemented pipeline (code in repository) contains the following components:

- `prepare_book.py` — remove boilerplate, produce cleaned book text.
- `character_extraction.py` — run multiple NER tools (spaCy, Stanza, GLiNER) to gather person mentions and sentence indices.

¹<https://networkofthrones.com/>

²<https://moviegalaxies.com/discover/movies/all>

³Vincent Labatut and Xavier Bost. 2019. Extraction and Analysis of Fictional Character Networks: A Survey. *ACM Computing Surveys* 52(5):89. <https://doi.org/10.1145/3344548>

- **cleaning** — normalize capitalization, strip punctuation, filter tokens via an ignore list.
- **alias resolution** — several methods (Levenshtein, SequenceMatcher, fuzzy/rapidfuzz); final greedy clustering using average similarity of the three metrics with a threshold of 85.
- **finalization** — build canonical character list (JSON) and sentence-indexed interactions.
- **interactions.py** — five methods: sliding-window heuristic; fastcoref coreference-based; Contextual Weighting with regex or with DL dialog attribution; local LLM analysis (Ollama with several models).

5 Methods (detailed)

5.1 Named Entity Recognition (NER)

We tested three NER systems:

- spaCy (pretrained English models e.g. `en_core_web_sm` / `en_core_web_lg`).
- Stanza (Stanford-style models).
- GLiNER (specialized lightweight model).

NER outputs are combined (set-union) to increase recall in candidate extraction.

5.2 Cleaning and normalization

NER outputs include noise (titles, punctuation, single-letter tokens). Cleaning consists of:

- punctuation stripping and case normalization,
- removing tokens or groups appearing in a manual ignore list, and a universal blacklist to remove locations or objects mistaken for people.
- deduplicating and sorting the candidate list.

5.3 Alias resolution

We explored several string-based similarity methods:

- Levenshtein-based similarity (custom implementation) calculates minimal edit distance between two names.
- SequenceMatcher (python difflib) measures character-level similarity ratios.
- Fuzzy matching (python fuzzywuzzy) provides token-aware similarity scores.

Finally, we applied a greedy clustering algorithm, which groups the closest name variants under a canonical name. It works as follows:

1. Names are sorted by frequency of occurrence in the text.
2. Each name is compared to all others using a combined similarity score (average of Levenshtein, Fuzzy, and SequenceMatcher)
3. Names exceeding a similarity threshold (85 out of 100) or contained within one another are clustered together.

5.4 Interaction detection

We tested five strategies:

1. **Sliding window + pronoun heuristics** — co-occurrence in the same line or close context + simple heuristic pronoun resolution using gender and recency.
2. **fastcoref-based coreference** — resolve mentions across sentences and map mentions to canonical character names
3. **Contextual Weighting:** Uses decaying weights and **Sentiment Analysis** (TextBlob) with regex-based **Speaker Detection**.
4. **Local LLM analysis (Ollama)** — prompt-based analysis per paragraph chunk; tested several models (Gemma, Qwen, Llama variants).
5. **Advanced Contextual:** Adds **Zero-Shot Classification** (BART) for deep dialogue attribution.

6 Evaluation protocol

For a small, human-curated test set ‘sample_gt.txt’ (22 ground-truth interactions — representing short excerpted paragraphs), we compared each method’s output against the ground truth.

- Matching is token-based / name-normalized (e.g. “Miss Elizabeth Bennet” and “Elizabeth” considered a match if tokens overlap in canonical mapping).
- Metrics reported: Precision, Recall, F1 computed over (line, character) pairs for the 22-line sample.
- This is an empirical comparison on the sample; full-book evaluation would require larger gold annotations.

Evaluation caveat. The numeric metrics presented in this report should be understood as estimates derived from a small manually annotated sample. They provide comparative insight into relative behaviour of methods on typical excerpts, but are not a comprehensive measure of performance on the whole book.

7 Results

7.1 Quantitative comparison (sample)

Table 1 shows the comparison on the sample (22 ground-truth items). These numbers are computed on the small sample provided in the repository (see Appendix).

Table 1: Comparison of methods on the sample dataset (22 ground-truth lines). Metrics computed over (line, character) pairs.

Method	Detected	TP	FP	FN	Precision	Recall
fastcoref	22	60	14	2	0.811	0.968
gemma	22	52	6	6	0.897	0.897
llama (local)	22	40	7	14	0.851	0.741
qwen (local)	21	47	4	9	0.922	0.839
sliding window	22	62	35	2	0.639	0.969

Numbers in the table are counts of matched (line,character) mentions; Precision/Recall are derived from those counts.

We summarize F1 (for convenience) in the bar chart below and show the number of detected interactions per method.

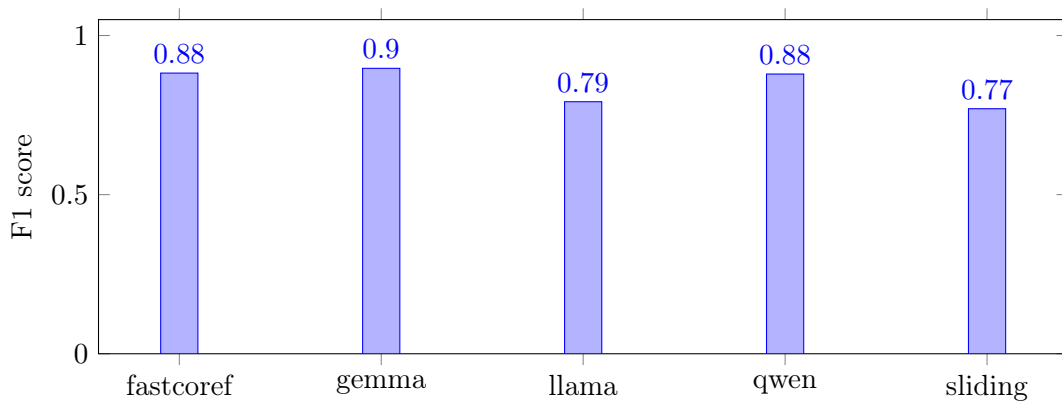


Figure 1: F1 (sample) per method.

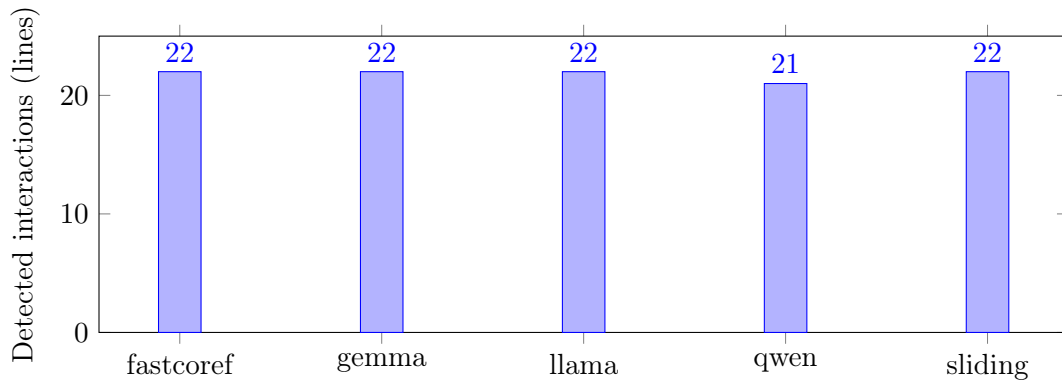


Figure 2: Number of detected interaction lines (sample).

7.2 Interpretation

- **Gemma:** best F1 on this sample (0.897). Balanced precision and recall: fewer false positives than sliding-window and fewer false negatives than some LLMs.
- **fastcoref:** very high recall (resolves many pronouns and mentions) but somewhat lower precision (some spurious linkings / place names interpreted as mentions).

- **Sliding-window heuristic:** very high recall but low precision: tends to aggregate many names appearing in context without real interaction (false positives).
- **LLM-based (local):** variable — Qwen performed well in precision, while the generic Llama variant missed more mentions on this sample.
- **Contextual Weighting:** Implemented a "memory" system where characters stay "active" in the context for several sentences, decaying over time. This captures interactions even when names aren't repeated in every line.
- **Speaker Detection:** Added logic to identify who is speaking in dialogues (e.g., "*said Elizabeth*"), ensuring the speaker is correctly included in the interaction.
- **Deep Learning Attribution:** Integrated a Transformers-based model (`facebook/bart-large-mnli`) for high-accuracy speaker identification in complex scenes.
- **Sentiment Analysis:** Every interaction now includes a **sentiment polarity score** (from -1.0 to +1.0), allowing to track the emotional tone of character relationships.

8 Why we chose Gemma (practical trade-off)

Our team tested three local LLM models and heuristics. The decision to prefer Gemma was driven by:

- **Empirical performance:** Best F1 on the small, human-annotated sample.
- **Speed / throughput:** In practice, processing the full book with chunking of 7 paragraphs per prompt on a GPU required 30–40 minutes (empirical measurement on our hardware). Gemma struck the best balance of accuracy and processing time among tested LLMs.
- **Stability and output formatting:** Gemma responses were more consistently formatted, lowering parser post-processing errors.

9 Qualitative analysis (typical errors)

Below are common error patterns observed across methods (examples taken from sample outputs and inspection):

False positives

- **Place names or objects recognized as persons:** e.g. *Pemberley Woods* or other named places were sometimes extracted as entities and included as interaction participants.
- **List aggregation in narrative:** sliding-window often treated lists of names in a descriptive sentence as interactions, even when those mentions are not interacting (e.g. a list of attendees).

False negatives

- **Title / full-name mismatches:** differences between canonical alias lists ("Miss Elizabeth Bennet" vs "Elizabeth") can cause misses unless aliases are normalized.
- **Coreference gaps:** where coreference resolution failed to map pronouns or nominal mentions to canonical names, interactions were missed.

Examples

- FP (fastcoref): detection of “Pemberley Woods” as participant (should be place).
Remedy: add a post-filter to remove known non-person entities (list of place-name heuristics or entity type checks).
- FN (some LLM chunks): splitting names into inconsistent forms (“Fitzwilliam Darcy” vs “Mr. Darcy”) — resolve via alias mapping and fuzzy matching.

10 Implementation notes and reproducibility

10.1 Environment

- Python 3.8+ recommended.
- Key libraries: `spacy`, `stanza`, `gliner` (or `gliner2` depending on package name), `rapidfuzz`, `fastcoref`, `requests`, `textblob`, `transformers` `tqdm`. Pin versions in `requirements.txt` for reproducibility.
- spaCy model: `python -m spacy download en_core_web_sm` (or `en_core_web_lg` if available).

10.2 Manual quality control

An important part of our workflow was manual inspection and correction of automatically produced candidate lists. The file `results/character_extraction/correct_character_list.txt` contains the curated list of canonical characters used for the final analyses. This step improved the quality of final outputs by resolving obvious false positives and standardising canonical name forms.

10.3 Data / outputs

Important output files (relative paths in repository):

- `results/character_extraction/final_characters.json` — canonical characters with aliases and sentence indices.
- `results/aliases/aliases_*.json` — alias groups produced by different algorithms.
- `results/interactions/pride_and_prejudice.txt` — full-book interaction outputs (one file per method).
- `results/character_extraction/correct_character_list.txt` — final manually-curated character list used for best results.

10.4 How to run (recommended order)

1. `pip install -r requirements.txt`
2. `python prepare_book.py`
3. `python character_extraction.py`
4. (Optional cleaning scripts: move / inspect `combined_characters.txt` and create `correct_character_list`)
5. `python interactions.py [method]` where method is:

- 1 = sliding window heuristic,
- 2 = fastcoref,
- 3 = contextual weighting
- 4 = LLM (Ollama)
- 5 = contextual weighting with dialog attribution.

11 Limitations and future work

- **Gold-standard evaluation:** current evaluation is on a small sample. A larger annotated dataset would allow robust precision/recall estimates.
- **Better coreference:** integrate more robust coreference models (e.g. span-based models from Hugging Face) or ensemble CR.
- **Dialogue detection:** detect speech/quotes to prioritize dialogue interactions over narrative co-occurrence.
- **Entity typing:** filter non-person entities (places, objects) earlier to reduce FP.
- **Name normalization with embeddings:** use name embeddings or contextual embeddings + clustering to improve alias resolution beyond string metrics.

12 Conclusions

We built a reproducible pipeline that extracts character candidates via multiple NER systems, canonicalises aliases and detects interactions at sentence level using several strategies. On a small hand-annotated sample, a local LLM model (Gemma) provided the best F1 / latency trade-off for interaction extraction; however, substantial improvements remain possible, especially regarding coreference and entity-type filtering.

A Appendix A: Sample gold / method outputs

Below are the short sample files used for evaluation (excerpts are exactly those provided in the repository). They are included so the reader can reproduce the evaluation in this report.

sample_gt.txt (gold)

1, Miss Elizabeth Bennet, Mr. Darcy
3, Miss Catherine Bennet, Miss Lydia Bennet, Mrs. Bennet
5, Miss Elizabeth Bennet, Miss Jane Bennet, Mr. Charles Bingley
5, Miss Caroline Bingley, Mr. Fitzwilliam Darcy
7, Miss Elizabeth Bennet, Mr. Fitzwilliam Darcy, Mr. George Wickham
9, Miss Elizabeth Bennet, Mr. Charles Bingley
11, Miss Elizabeth Bennet, Mr. Darcy
13, Miss Elizabeth Bennet, Mr. Darcy
13, Miss Jane Bennet, Mr. Charles Bingley
13, Mrs. Bennet, Miss Elizabeth Bennet, Mr. Darcy, Miss Jane Bennet, Mr. Charles Bingley, Miss
15, Miss Caroline Bingley, Miss Elizabeth Bennet, Mr. Darcy
17, Miss Jane Bennet, Miss Elizabeth Bennet, Mr. Charles Bingley, Mr. Darcy
19, Mrs. Bennet, Mr. Collins, Miss Catherine Bennet
21, Miss Elizabeth Bennet, Mr. Darcy, Mr. Charles Bingley
23, Mr. George Wickham, Miss Elizabeth Bennet
25, Miss Lydia Bennet, Miss Catherine Bennet, Colonel Forster, Captain Carter
27, Miss Elizabeth Bennet, Mr. Darcy
29, Miss Elizabeth Bennet, Mr. Darcy
31, Mrs. Bingley, Mr. Charles Bingley
33, Miss Lydia Bennet, Miss Mary Bennet
35, Mr. Collins, Mrs. Bennet, Miss Charlotte Lucas
37, Miss Jane Bennet, Mr. Charles Bingley
39, Miss Elizabeth Bennet, Mrs. Gardiner
41, Colonel Fitzwilliam, Miss Elizabeth Bennet, Mr. Darcy
43, Lady Catherine de Bourgh, Miss Elizabeth Bennet, Rev. William Collins

sample_fastcoref.txt (method output)

1, Darcy, Elizabeth
3, Bennet, Kitty, Lydia
5, Bingley, Caroline, Darcy, Elizabeth, Jane, Mr. Bingley
7, Darcy, Elizabeth, Wickham
9, Bennet, Bingley, Elizabeth, Mr. Bingley
11, Darcy, Elizabeth, Pemberley Woods
13, Bennet, Bingley, Darcy, Elizabeth, Georgiana, Jane
15, Bingley, Caroline, Darcy, Elizabeth
17, Bennet, Bingley, Darcy, Elizabeth, Miss King, Mr. Bingley
19, Bennet, Collins, De Bourgh, Denny, Lady Catherine, Rosings
21, Bingley, Darcy, Elizabeth
23, Elizabeth, Wickham
25, Captain, Carter, Colonel, Forster, Kitty, Lydia
27, Darcy, Elizabeth
29, Darcy, Elizabeth
31, Bingley, Mr. Bingley
33, Lydia, Mary

35, Bennet, Charlotte, Collins
37, Bingley, Collins, Jane, Mr. Bingley
39, Elizabeth, Gardiner
41, Colonel, Darcy, Elizabeth, William
43, Collins, De Bourgh, Denny, Elizabeth, Lady Catherine

sample_gemma.txt (method output)

1, Miss Elizabeth Bennet, Mr. Darcy
3, Kitty, Lydia Wickham, Mrs. Bennet
5, Caroline Bingley, Mr. Darcy
5, Miss Elizabeth Bennet, Miss Jane Bennet
5, Miss Elizabeth Bennet, Miss Jane Bennet, Mr. Bingley
5, Miss Elizabeth Bennet, Miss Jane Bennet, Mr. Charles Bingley
7, Miss Elizabeth Bennet, Mr. Darcy, Mr. Wickham
9, Miss Elizabeth Bennet, Miss Jane Bennet, Mr. Charles Bingley, Mrs. Bennet
11, Miss Elizabeth Bennet, Mr. Darcy
13, Elizabeth Bennet, Georgiana Darcy, Jane Bennet, Mr. Darcy, Mrs. Bennet
13, Elizabeth Bennet, Mr. Darcy
13, Jane Bennet, Mr. Charles Bingley
15, Caroline Bingley, Elizabeth Bennet, Mr. Darcy
17, Elizabeth, Miss Bennet, Mr. Bingley, Mr. Darcy
19, Mr. Collins, Mrs. Bennet
21, Mr. Charles Bingley, Mr. Darcy
23, Elizabeth, Mr. Wickham
25, Captain Carter, Colonel Forster, Miss Catherine Bennet, Miss Lydia Bennet
27, Miss Elizabeth Bennet, Mr. Darcy
29, Miss Elizabeth Bennet, Mr. Darcy
31, Mr. Bingley, Mrs. Bennet
33, Miss Lydia Bennet, Miss Mary Bennet, Mrs. Bennet
35, Charlotte, Mr. Collins, Mrs. Bennet
37, Miss Jane Bennet, Mr. Bingley
37, Mr. Bingley, Mr. Collins, Mr. Phillips
39, Miss Elizabeth Bennet, Mrs. Gardiner
41, Colonel Fitzwilliam, Miss Elizabeth Bennet, Mr. Darcy
43, Lady Catherine de Bourgh, Miss Elizabeth Bennet
43, Lady Catherine de Bourgh, Mr. Collins

sample_llama.txt (method output)

1, Darcy, Elizabeth
3, Kitty, Mrs. Bennet
5, Elizabeth, Jane
7, Darcy, Wickham
9, Mr. Bingley, Mrs. Bennet
11, Darcy, Elizabeth
13, Darcy, Elizabeth
15, Caroline Bingley, Elizabeth
17, Miss Bennet, Mr. Bingley
19, Mr. Collins, Mrs. Bennet
21, Bingley, Mr. Darcy
23, Elizabeth, Mr. Wickham

25, Captain Carter, Colonel Forster, Kitty, Lydia
27, Darcy, Elizabeth
29, Elizabeth, Mr. Darcy
31, Mr. Bingley, Mrs. Bennet
33, Lydia, Mary
35, Charlotte, Mr. Collins
37, Mr. Bingley, Mr. Collins, Mr. Phillips
39, Elizabeth, Mrs. Gardiner
41, Elizabeth, Mr. Darcy
43, Elizabeth, Lady Catherine de Bourgh

sample_qwen.txt (method output)

1, Miss Elizabeth Bennet, Mr. Darcy
3, Miss Catherine Bennet, Miss Lydia Bennet, Mrs. Bennet
3, Miss Catherine Bennet, Mrs. Bennet
5, Miss Caroline Bingley, Mr. Fitzwilliam Darcy
5, Miss Elizabeth Bennet, Miss Jane Bennet
5, Miss Elizabeth Bennet, Miss Jane Bennet, Mr. Charles Bingley
7, Miss Elizabeth Bennet, Mr. Fitzwilliam Darcy, Mr. George Wickham
9, Miss Elizabeth Bennet, Mr. Charles Bingley
9, Mr. Charles Bingley, Mrs. Bennet
11, Miss Elizabeth Bennet, Mr. Darcy
13, Miss Elizabeth Bennet, Mr. Darcy
13, Miss Jane Bennet, Mr. Charles Bingley
15, Miss Caroline Bingley, Miss Elizabeth Bennet, Mr. Darcy
17, Miss Elizabeth Bennet, Miss Jane Bennet
17, Miss Elizabeth Bennet, Mr. Darcy
17, Miss Jane Bennet, Mr. Charles Bingley
19, Mr. Collins, Mrs. Bennet
21, Mr. Charles Bingley, Mr. Darcy
23, Miss Elizabeth Bennet, Mr. George Wickham
25, Captain Carter, Colonel Forster, Miss Catherine Bennet, Miss Lydia Bennet
27, Miss Elizabeth Bennet, Mr. Darcy
29, Miss Elizabeth Bennet, Mr. Fitzwilliam Darcy
33, Miss Lydia Bennet, Miss Mary Bennet, Mrs. Bennet
35, Miss Charlotte Lucas, Mr. Collins
35, Mr. Collins, Mrs. Bennet
37, Miss Jane Bennet, Mr. Charles Bingley
37, Mr. Charles Bingley, Mr. Collins, Mr. Phillips
39, Miss Elizabeth Bennet, Mrs. Gardiner
41, Colonel Fitzwilliam, Miss Elizabeth Bennet, Mr. Darcy
43, Lady Catherine de Bourgh, Miss Elizabeth Bennet, Rev. William Collins

sample_sliding_window.txt (method output)

1, Darcy, Elizabeth
3, Bennet, Darcy, Elizabeth, Kitty, Lydia
5, Bennet, Bingley, Caroline, Darcy, Elizabeth, Jane, Kitty, Lydia, Mr. Bingley
7, Darcy, Elizabeth, Wickham
9, Bennet, Bingley, Elizabeth, Mr. Bingley, Wickham
11, Bingley, Darcy, Elizabeth, Pemberley Woods

13, Bennet, Bingley, Darcy, Elizabeth, Georgiana, Jane
 15, Bingley, Caroline, Darcy, Elizabeth
 17, Bennet, Bingley, Darcy, Elizabeth, Miss King, Mr. Bingley
 19, Bennet, Bingley, Collins, De Bourgh, Denny, Elizabeth, Lady Catherine, Rosings
 21, Bennet, Bingley, Darcy, Elizabeth, Lady Catherine
 23, Bingley, Elizabeth, Wickham
 25, Captain, Carter, Colonel, Elizabeth, Forster, Kitty, Lydia, Wickham
 27, Darcy, Elizabeth, Kitty
 29, Darcy, Elizabeth
 31, Bingley, Darcy, Mr. Bingley
 33, Elizabeth, Lydia, Mary
 35, Bennet, Bingley, Charlotte, Collins, Lydia
 37, Bingley, Collins, Jane, Mr. Bingley
 39, Collins, Elizabeth, Gardiner, Jane
 41, Collins, Colonel, Darcy, Elizabeth, William
 43, Collins, De Bourgh, Denny, Elizabeth, Lady Catherine, William

`sample_contextual_weighting_heuristic.txt(methodoutput)`

5, Bennet, Mrs. Long
 6, Bennet, Mrs. Long
 7, Bennet, Mrs. Long
 8, Bennet, Mrs. Long
 10, Morris, Mrs. Long
 11, Bingley, Morris, Mrs. Long
 12, Bingley, Morris, Mrs. Long
 13, Bingley, Morris, Mrs. Long
 14, Bennet, Bingley
 28, Lucas, William
 29, Lucas, William
 30, Lucas, William
 31, Bingley, Lizzy, Lucas, William
 32, Bingley, Jane, Lizzy, Lydia
 33, Bingley, Jane, Lizzy, Lydia
 34, Bingley, Jane, Lizzy, Lydia
 35, Bennet, Jane, Lizzy, Lydia
 36, Bennet, Lizzy
 37, Bennet, Lizzy
 51, Bennet, Bingley
 52, Bennet, Bingley
 53, Bennet, Bingley
 54, Bennet, Bingley, Lizzy
 55, Bingley, Lizzy
 56, Bingley, Elizabeth, Lizzy, Mamma, Mrs. Long
 57, Bingley, Elizabeth, Lizzy, Mamma, Mrs. Long
 58, Bingley, Elizabeth, Mamma, Mrs. Long
 59, Elizabeth, Mamma, Mrs. Long
 60, Bennet, Elizabeth, Mrs. Long
 62, Bennet, Kitty
 63, Bennet, Kitty
 64, Bennet, Kitty
 66, Kitty, Lizzy

67, Kitty, Lizzy, Mrs. Long
 68, Bennet, Bingley, Kitty, Lizzy, Mrs. Long
 69, Bennet, Bingley, Kitty, Lizzy, Mrs. Long
 70, Bennet, Bingley, Mrs. Long
 71, Bennet, Bingley, Mrs. Long
 73, Bennet, Mrs. Long
 74, Bennet, Mrs. Long
 77, Bennet, Mary
 80, Bingley, Mary
 81, Bingley, Mary
 82, Bingley, Mary
 83, Bingley, Mary
 90, Bennet, Kitty

`sample_contextualwweightingDL.txt(methodoutput)`

5, Bennet, Mrs. Long
 6, Bennet, Mrs. Long
 7, Bennet, Mrs. Long
 8, Bennet, Mrs. Long
 10, Bennet, Morris, Mrs. Long
 11, Bingley, Morris, Mrs. Long
 12, Bingley, Morris, Mrs. Long
 13, Bingley, Morris, Mrs. Long
 14, Bennet, Bingley
 28, Bingley, Lucas, William
 29, Lucas, William
 30, Lucas, William
 31, Bingley, Lizzy, Lucas, William
 32, Bingley, Jane, Lizzy, Lydia
 33, Bingley, Jane, Lizzy, Lydia
 34, Bingley, Jane, Lizzy, Lydia
 35, Bennet, Jane, Lizzy, Lydia
 36, Bennet, Lizzy
 37, Bennet, Lizzy
 51, Bennet, Bingley
 52, Bennet, Bingley
 53, Bennet, Bingley
 54, Bennet, Bingley, Lizzy
 55, Bingley, Lizzy
 56, Bingley, Elizabeth, Lizzy, Mamma, Mrs. Long
 57, Bingley, Elizabeth, Lizzy, Mamma, Mrs. Long
 58, Bingley, Elizabeth, Mamma, Mrs. Long
 59, Elizabeth, Mamma, Mrs. Long
 60, Bennet, Elizabeth, Mrs. Long
 62, Bennet, Kitty
 63, Bennet, Kitty
 64, Bennet, Kitty
 66, Kitty, Lizzy
 67, Kitty, Lizzy, Mrs. Long
 68, Bennet, Bingley, Kitty, Lizzy, Mrs. Long
 69, Bennet, Bingley, Kitty, Lizzy, Mrs. Long

70, Bennet, Bingley, Mrs. Long
71, Bennet, Bingley, Mrs. Long
73, Bennet, Mrs. Long
74, Bennet, Mrs. Long
77, Bennet, Mary
80, Bingley, Mary
81, Bingley, Mary
82, Bingley, Mary
83, Bingley, Mary
84, Bennet, Bingley
90, Bennet, Kitty

B Appendix B: Files produced by pipeline (short guide)

- `results/character_extraction/cleaned_characters.txt` — cleaned candidate names.
- `results/character_extraction/combined_characters.txt` — union of NER outputs.
- `results/character_extraction/final_characters.json` — final canonical characters (aliases + positions).
- `results/aliases/aliases_*.json` — alias groups using different similarity metrics.
- `results/interactions/pride_and_prejudice_*.txt` — full-book interactions outputs for each method.